

## **2. REQUIREMENT ANALYSIS PHASE**

Project: EduGenie – Google Gemini Powered Learning Assistant

### **2.1 Introduction**

Requirement Analysis is an important phase in the software development life cycle. It focuses on identifying, understanding, documenting, and organizing the requirements of a proposed system. For the EduGenie project, this phase converts the ideas identified during brainstorming into clear system requirements that can guide design and implementation.

The main purpose of requirement analysis is to understand what the users expect from the system and what the system must provide. It also helps the development team define the scope of the project, identify technical needs, and reduce confusion during later stages.

### **2.2 Purpose of Requirement Analysis**

- Identify the needs of students who will use EduGenie.
- Define the functions that the system should provide.
- Identify technical and software requirements.
- Specify quality requirements such as usability and reliability.
- Understand project limitations and constraints.
- Create a clear foundation for system design and development.

### **2.3 Requirement Analysis in EduGenie**

EduGenie is planned as a Google Gemini powered learning assistant. Therefore, the requirement analysis considers both educational requirements and artificial-intelligence-based interaction. The system should allow users to enter learning-related questions and receive understandable responses. The requirements also consider the need for a simple interface and responsible use of AI-generated information.

## **2.4 Requirement Identification**

### **Understanding User Needs**

The first step in requirement analysis is identifying the people who will interact with the system and understanding their expectations. The primary users of EduGenie are students who require additional support during learning, revision, homework preparation, and self-study.

### **User Needs Identified**

- Students need a simple way to ask academic questions.
- They need explanations that are easy to understand.
- They may require examples to clarify difficult concepts.
- They need quick access to learning assistance.
- They may want to continue a topic through follow-up questions.
- They need an interface that does not require advanced technical knowledge.

### **System Needs**

The system must provide an interface through which the user can enter a question or request. The request must be processed and sent to the AI service through the selected integration. The generated response must then be displayed to the user in a readable form.

### **Scope Identification**

The initial scope of EduGenie is focused on educational assistance rather than attempting to replace teachers or classroom learning. The project concentrates on conversational academic support and can be expanded with additional features in future versions.

### **Stakeholder Consideration**

Students are the main stakeholders, while project team members, mentors, and educational institutions may also be interested in the system. Their expectations may differ, so the requirements are organized to balance usability, educational value, technical feasibility, and project limitations.

## **2.5 Functional Requirements**

### **Meaning of Functional Requirements**

Functional requirements describe what the system should do. They define the services, operations, and interactions expected from EduGenie. These requirements form the core of the proposed application.

#### **FR-01: User Input**

The system should provide a suitable input area where students can enter questions, doubts, or learning requests.

#### **FR-02: Request Processing**

The system should receive the user's request and process it through the configured AI service. The request should be handled in a structured manner.

#### **FR-03: AI-Generated Response**

The system should return an educational response generated using the integrated Google Gemini capability. The response should be displayed clearly to the user.

#### **FR-04: Conversational Interaction**

The system should support follow-up questions so that a student can continue discussing a topic rather than starting a completely new interaction each time.

#### **FR-05: Explanation Support**

The system should be capable of providing explanations, examples, summaries, or other educational assistance according to the user's request.

#### **FR-06: Error Handling**

The system should provide an appropriate message when an input is invalid, the service is temporarily unavailable, or a request cannot be processed.

#### **FR-07: Response Display**

Generated responses should be presented in a readable and organized format so that students can easily understand the information.

#### **FR-08: Future Extension**

The design should allow additional educational functions to be integrated later, such as quizzes, topic-based practice, or personalized learning support.

## **2.6 Non-Functional Requirements**

### **Meaning of Non-Functional Requirements**

Non-functional requirements describe the quality and operational characteristics of the system. They do not directly define a specific function, but they influence how effectively and comfortably users can use EduGenie.

### **Usability**

The interface should be simple, clear, and easy to understand. A student should be able to enter a question and view the response without unnecessary navigation.

### **Performance**

The system should process normal requests within a reasonable amount of time. Response performance may depend on network conditions and the external AI service.

### **Reliability**

The application should handle normal user interactions consistently. Where an external service is unavailable, the system should respond with a suitable error message instead of failing without explanation.

### **Maintainability**

The project should be organized so that developers can understand and modify the code. Components should be kept logically separated wherever possible.

### **Scalability**

The system should have a structure that permits future improvements and additional learning features without requiring a complete redesign.

### **Security and Responsible Use**

The application should avoid unnecessary collection of personal information. API credentials and sensitive configuration details should not be exposed in publicly accessible source code. Users should also be reminded that AI-generated educational information may require verification.

### **Compatibility**

The system should work with the selected development environment and the devices or browsers targeted by the project. Compatibility requirements should be tested during later implementation.

## 2.7 User Requirements

### Primary User: Student

The primary user requirement is that students should be able to interact with EduGenie easily. The application should support students who have different levels of subject knowledge and technical familiarity.

### Basic User Requirements

- The user should be able to access the learning assistant.
- The user should be able to enter a question or learning request.
- The user should be able to submit the request.
- The user should be able to read the generated response.
- The user should be able to ask follow-up questions.
- The user should receive understandable feedback when an error occurs.

### Learning-Oriented Requirements

Students may use the system for concept clarification, revision, examples, summaries, and general study assistance. The system should respond according to the context of the request as far as the integrated AI capability permits.

### Ease of Use

The interface should avoid unnecessary complexity. Labels, buttons, text fields, and response areas should have clear meanings. The user should not need advanced programming knowledge to operate the learning assistant.

### Accessibility Considerations

The application should use readable text, clear spacing, and understandable interaction elements. Where possible, the design should support comfortable use on commonly available devices.

### User Expectations

Users generally expect responses to be relevant, understandable, and reasonably quick. Since generative AI can occasionally produce inaccurate information, the system should be treated as an assistance tool and important academic information should be verified using appropriate sources.

## **2.8 Hardware and Software Requirements**

### **Hardware Requirements**

- Computer or laptop for development and testing.
- Keyboard and mouse or equivalent input devices.
- Sufficient RAM and storage for the development environment.
- Stable internet connection for cloud-based AI interaction and related services.
- Mobile or desktop device for user-interface testing, where applicable.

### **Software Requirements**

- Operating system compatible with the selected development tools.
- Web browser for application access and testing.
- Code editor or integrated development environment.
- Required programming language and runtime environment.
- Google Gemini / applicable AI integration service.
- Version-control platform such as GitHub for project management.
- Document and presentation software for project documentation.

### **Development Environment**

The exact development environment may depend on the implementation approach selected by the project team. The environment should support application development, API integration, debugging, testing, and version control.

### **Internet and API Requirement**

Because EduGenie uses an AI service, network connectivity is an important requirement. The application may need access to the relevant Gemini service through an appropriate API or integration method. Authentication credentials should be stored securely and should not be included directly in publicly shared source files.

### **Documentation Requirement**

The project should maintain documentation describing requirements, system design, implementation, testing, and results. Proper documentation helps the team explain the project during evaluation and supports future maintenance.

## **2.9 Feasibility, Constraints and Requirement Prioritization**

### **Technical Feasibility**

The proposed requirements are technically feasible using modern software development tools and generative AI services. However, implementation depends on the selected technology stack, available APIs, internet connectivity, and the team's technical skills.

### **Operational Feasibility**

EduGenie is intended to be simple enough for students to operate. The conversational nature of the system can make educational interaction more accessible, while the project scope keeps the initial implementation manageable.

### **Schedule Feasibility**

The requirements were considered with the available academic project timeline in mind. Core features should be implemented first, while optional features can be considered only if sufficient time remains.

### **Project Constraints**

- Dependence on internet connectivity and external AI services.
- Possible limitations of API usage, service availability, or quotas.
- Limited development time and team resources.
- AI responses may occasionally be inaccurate or incomplete.
- Project scope must remain manageable for successful completion.

### **Requirement Prioritization**

Requirements can be grouped into essential and optional categories. Essential requirements include user input, request processing, AI response generation, response display, basic error handling, and a usable interface. Optional future requirements may include quizzes, personalized study plans, advanced analytics, or additional educational modules.

### **Importance of Prioritization**

Prioritization helps the team focus on the minimum functional product first. It prevents the project from becoming unnecessarily large and ensures that the core educational purpose remains clear.

## 2.10 Requirement Analysis Conclusion

### Summary

The Requirement Analysis Phase converted the EduGenie concept into a structured set of user, functional, non-functional, technical, and operational requirements. This phase provides the development team with a clear understanding of what the system should achieve and what resources are required.

### Key Outcomes

- Primary users and their learning needs were identified.
- Functional requirements for the learning assistant were defined.
- Non-functional requirements such as usability, reliability, performance, and maintainability were identified.
- Hardware, software, internet, and AI-service requirements were documented.
- Project constraints and possible risks were considered.
- Core requirements were prioritized for implementation.
- Future extension possibilities were identified.

### Importance for the Next Phase

The documented requirements will serve as the foundation for the system design phase. The design team can use them to plan the system architecture, user interface, data flow, AI integration, and other technical components. Testing criteria can also be derived from these requirements.

### Final Statement

A well-defined requirement analysis reduces misunderstanding between the project objectives and the final implementation. For EduGenie, it establishes a clear direction for developing an interactive AI-powered learning assistant. By keeping the core requirements focused on student needs, the project can proceed to design and implementation in a systematic manner.

### Transition to System Design

After completing requirement analysis, the next stage is to convert these requirements into a detailed system design. The design phase will define how the identified functions are organized, how users interact with the application, and how the Gemini-powered response process is integrated into the overall system.